Retrieval Grounding, Claim Verification and Abstention for a Trustworthy Library Assistant

Project Report - Hallucination and Knowledge-Cutoff Probe for a Library Assistant

Abstract - Detecting hallucinations is only useful if the assistant can act on the result. This paper describes a grounding-and-verification architecture for a library assistant that combines retrieval-augmented generation (RAG) over the library catalogue, a post-generation claim verifier, and an abstention policy driven by the probe results of Papers 1 and 2. We outline the pipeline, a verification procedure using sampling-consistency and catalogue lookup, an evaluation design with ablations, and an error-analysis plan. The aim is to reduce phantom-item and stale-information errors while preserving helpfulness.

Keywords: retrieval-augmented generation, verification, abstention, hallucination mitigation, evaluation

I. Introduction

Papers 1 and 2 define what to measure: five hallucination types and four temporal probe families. Here we ask what to *do* about the failures. Retrieval-augmented generation grounds responses in external documents and is the standard mitigation for knowledge-intensive tasks [1], [2]. In a library setting the retrievable corpus is natural: the catalogue, the policy pages and the new-acquisitions feed. Yet retrieval alone does not guarantee faithfulness, since models may ignore or over-extend retrieved evidence [3]. We therefore add a verification stage and an explicit abstention rule.

II. System Architecture

The pipeline has four stages. (1) **Query classifier:** labels the query as bibliographic, policy, availability, or general reference, and flags time-sensitivity using the T1-T4 families. (2) **Retriever:** queries a catalogue search index (exact match on ISBN, title and author, plus dense semantic search) and a policy-page index. (3) **Generator:** an LLM instructed to answer only from retrieved records, to quote identifiers verbatim, and to cite the record ID for every claim. (4) **Verifier:** checks each generated claim before the response is shown.

Identifiers are never generated freely. ISBNs, call numbers and DOIs are copied from retrieved fields, and any identifier in the output that is absent from the retrieved set is removed or flagged. This single rule is designed to eliminate most H2 and H3 errors by construction.

III. Verification and Abstention

The verifier applies two complementary checks. **Catalogue lookup** verifies named entities and identifiers by exact search, which is cheap and precise for H1, H2 and H5. **Sampling consistency** follows SelfCheckGPT [4]: several additional responses are sampled and claims that are not consistently reproduced are treated as likely hallucinations, which helps for free-text claims without a catalogue entry. Claims are decomposed into atomic statements, as in FActScore [5], so the response can be partially corrected rather than discarded.

The **abstention policy** uses three signals: retrieval returned nothing relevant; the verifier rejected a core claim; or the query falls in a time-sensitive family near or beyond the effective cutoff found in Paper 2. If any fires, the assistant says it could not confirm the information and offers a catalogue link or a librarian contact. Because models carry partial knowledge of their own uncertainty [6], model confidence may be used as a fourth, optional signal, but is not trusted alone.

IV. Evaluation Design

We evaluate on the probe sets from Papers 1 and 2 using four system variants:

Variant	Components	Purpose
S0	LLM only	Baseline hallucination and staleness
S1	S0 + retrieval	Effect of grounding

Variant	Components	Purpose
S2	S1 + identifier rule + verifier	Effect of verification
S3	S2 + abstention policy	Full system

Primary measures are per-type hallucination rate, claim-level factual precision, TSR and CAS. Because mitigation can make an assistant overly cautious, we also report *answer rate* on answerable queries and *abstention precision* on unanswerable ones, and we show them together so that gains are not obtained by refusing. A small human study with librarians rates usefulness and tone.

V. Error Analysis and Limitations

Expected residual failures include retrieval misses (the book exists but the search did not find it), faithful-but-wrong answers caused by erroneous catalogue records, and verifier false positives that suppress correct claims. Each failure is categorised and fed back into probe revision. The architecture adds latency from retrieval and sampling; a cost-latency profile is therefore reported alongside accuracy. The design assumes a reasonably clean catalogue and API access, which smaller libraries may lack.

VI. Conclusion

We described a retrieval-grounded, verified and abstention-aware library assistant whose behaviour is measured with the taxonomy of Paper 1 and the temporal probes of Paper 2. The ablation design lets a library see which component contributes most, and the abstention analysis keeps the focus on trustworthy rather than merely fluent answers.

References

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